

SAMPLE CAPABILITY INDICES: DISTRIBUTIONAL PROPERTIES AND SOME APPLICATIONS

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Summary: Capability indices (CIs) are popular in the industry, typically used to evaluate a process under statistical control. Being functions of the process mean and the standard deviation, the CIs are actually unknown parameters and yet, in the past, point estimates of these indices, obtained from sampled data, were often used to make the decision, ignoring their sampling variability. In this paper we focus on the distributional and inferential properties of the sample estimators of four popular capability indices C_P , C_{PU} , C_{PL} and C_{Pk} and show how the exact sampling distributions of the estimated CIs can be derived using the method of conditioning. Important attributes of that distribution, such as the mean and the standard deviation, can then be calculated to arbitrary precision. Hence these important results become more accessible to the practitioners engaged in real-world applications. We also develop hypothesis testing and confidence interval estimation of the CIs,

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since some of the existing methodology based on a normal approximation is seen to be questionable. Instead, we propose small sample inference based on our distributional results. Finally, we emphasize and illustrate the fact that neither C_P nor C_{Pk} , alone, can adequately reflect the capability of a process. We propose making joint inference on C_P and C_{Pk} , using a confidence region that follows from the exact joint distribution of the estimated CIs derived here.

1. Introduction

Statistical process control (SPC) techniques are employed to establish that a process operates at or near a target, subject to random variation. Once statistical control is confirmed, the need arises to ascertain whether or not the process is capable of producing items within the specified tolerance. Capability indices (CIs) are used at this stage to help evaluate this property of the process. Some of the best known CIs are C_P , C_{PU} , C_{PL} and C_{Pk} . The literature contains excellent expositions of the practical value and uses of these CIs (see, for example, Kotz & Johnson (2002) and Spiring et al. (2003)). Suffice it to add here, that in a Six-sigma program, the practitioner will typically use information contained in the capability indices to assess and monitor the improvement made in processes under study. Note, however, that these indices are functions of (unknown) population parameters such as the mean and the standard deviation and are therefore (unknown) parameters themselves. In the past, it was not uncommon for users to substitute sample estimates (of the underlying parameters) in these formulae to calculate the indices and use the final result as if they were the true values, completely ignoring the sampling variation of the estimators. Since the late 1980s researchers seem to recognize that the estimated CIs are random variables and began studies of the sampling distributions of these estimators. To this end some distributional

theory has been developed in the literature and both exact and approximate results have been presented. It is generally agreed that the distributional properties of at least some capability indices involve complicated mathematics and cumbersome computations. This is possibly one reason why some of the results in literature are either based on approximations or tables of limited extent. Such complications, we surmise, might also be partly responsible for these results to be under-utilized by the practitioners.

Aim of this paper

- The aim of this paper is to show how the exact distribution of some of these estimated capability indices can be derived by using the so-called *conditioning approach*. With this approach not only can we find distributional and numerical results more directly (that were previously produced by complex computer programming), but we can also obtain valuable insight and demonstrate some shortcomings in the tables and conclusions reached earlier (through approximation based approaches) in the literature.
- Furthermore, small sample statistical inference regarding these indices becomes either more direct to conduct (despite what was previously found by other authors – see Vännman (1997)) or even possible for the first time. One specific problem we address concerns the fact that the distribution of the estimator of C_{P_k} depends not only on the parameter (the capability index C_{P_k}) itself but also on a nuisance parameter (another capability index, C_P). We propose to overcome this problem by implementing joint inference on both indices. All results are illustrated with examples.

2. Literature overview

Before we discuss some of the related literature, we first define the four CIs mentioned above and their natural estimators. Although there are many more CIs available in the literature (see for example the review by Kotz & Johnson, 2002, and the responses) we consider these four popular indices as illustrations

of our approach.

$$\begin{aligned}
 C_P &= \frac{U - L}{6\sigma} & \text{and} & \quad \hat{C}_P = \frac{U - L}{6S} \\
 C_{PU} &= \frac{U - \mu}{3\sigma} & \text{and} & \quad \hat{C}_{PU} = \frac{U - \bar{X}}{3S} \\
 C_{PL} &= \frac{\mu - L}{3\sigma} & \text{and} & \quad \hat{C}_{PL} = \frac{\bar{X} - L}{3S} \\
 C_{Pk} &= \min(C_{PU}, C_{PL}) & \text{and} & \quad \hat{C}_{Pk} = \min(\hat{C}_{PU}, \hat{C}_{PL})
 \end{aligned}$$

where U = upper specification limit, L = lower specification limit, μ and σ are the process mean and standard deviation, and $\bar{X}$ and S are the mean and the standard deviation, respectively, of a reference sample (obtained from a process in statistical control).

The C_P -index is used to compare the actual process variation to the allowable spread or variation. Larger values of the C_P -index indicate a higher conformance rate. Since C_P does not use (or give) information regarding the centre of the process, other indices are often used. Among these, the more popular ones are C_{PU} and C_{PL} when one-sided tolerances are specified and the C_{Pk} when two-sided tolerances are specified.

Since the sample capability indices are estimates and hence are random variables, their sampling distributions are essential for making valid statistical inference based on the indices, say, in the form of a test of hypothesis or a confidence interval. These are crucial in practical applications. To this end, Chou and Owen (1989) present expressions for the exact distributions of the estimators of all four CIs given above, assuming that the process distribution is normal. The normal distribution remains as one of the most common distributional assumptions in the quality-capability literature. Specifically,

they use the fact that under normality, the distribution of $\hat{C}_P$ is inversely proportional to a chi-distribution (the square root of a chi-square distribution) to obtain the probability density function (pdf) of $\hat{C}_P$ and further obtain the expected value and variance of the estimator. For the estimators $\hat{C}_{PU}$ and $\hat{C}_{PL}$, they show that both have distributions that are proportional to non-central t -distributions, from which the associated density functions and moment expressions are derived. Finally, Chow and Owen derived the cumulative distribution function (cdf), the probability density function (pdf) and the first two moments of $\hat{C}_{Pk}$, which are fairly complicated expressions, all of which are functions of two capability index parameters (C_{PU} and C_{PL} or C_P and C_{Pk}).

Chou, Owen, Borrego (1990) derived expressions for the lower confidence limits of the four capability indices mentioned above. In the case of C_{PU} , C_{PL} and C_{Pk} , these expressions are based on the non-central t -distribution and they produced tables from which the practitioner can find the lower limits for a range of sample sizes and observed (sample) estimates of the indices. However, it is not fully clear how the confidence limits for C_{Pk} were derived. We maintain that some additional information is necessary in order to derive the confidence limits for C_{Pk} , since the distribution of $\hat{C}_{Pk}$ depends on two parameters (e.g. C_{PU} and C_{PL} or C_P and C_{Pk} – see our derivations below). On this point, also see the discussion in Kushler and Hurley (1992).

Lin and Sheen (2005), for example, acknowledge the fact that C_{Pk} depends on μ and σ and state that the position of the process mean μ relative to $M = \frac{L+U}{2}$ needs to be known to overcome the problem of dependence on two parameters. They also point out, as do many others in the literature, that using the non-central t -distribution is difficult and not popular in quality engineering.

Li, Owen and Borrego (1990), in deriving an expression for a confidence statement regarding C_{PU} (or C_{PL}), assume $U = \bar{X} + k_2 \bar{R}/d_2$ and $L = \bar{X} - k_2 \bar{R}/d_2$, where $\bar{R}/d_2$ is an unbiased estimator of σ based on the sample range R and $k_2 = 3\hat{C}_{PU}$. However, this implies assuming $\bar{X} = \frac{U+L}{2}$, which does not seem practical.

Guirguis and Rodriguez (1992) consider various approaches to find a confidence interval for C_{Pk} (both exactly and approximately). They provide a Fortran program by which the confidence interval for C_{Pk} can be derived, using the formulas in Chou and Owen (1990), but also under additional assumptions. Furthermore, they give a Fortran program to derive the pdf of $\hat{C}_{Pk}$. This density function is graphed and compared to the standard normal distribution for various choices of parameter values to provide support for the assumption that the distribution of $\hat{C}_{Pk}$ can be approximated by the normal distribution. Our observations, however, lead to questioning this assumption. See discussions later.

Kotz and Johnson (1993, pp. 55–65), give expressions for $E(\hat{C}_{Pk})$ and $Var(\hat{C}_{Pk})$ under normality and calculate some numerical values for various combinations of the parameter values. From these they conclude that for sample sizes as large as 40 there is still substantial uncertainty in the estimator $\hat{C}_{Pk}$ and it would be unwise to regard $\hat{C}_{Pk}$ as the true value of C_{Pk} .

Chou and Polansky (1993) use the exact distributions of $\hat{C}_{PU}$ and $\hat{C}_{PL}$ (related to the non-central t -distribution) to derive critical values for testing

	$H_0 : C_{PL} \leq C_{PL}(0)$	(the process is not capable)
against	$H_a : C_{PL} > C_{PL}(0)$	(the process is capable)
and for	$H_0 : C_{PU} \leq C_{PU}(0)$	(the process is not capable)
against	$H_a : C_{PU} > C_{PU}(0)$	(the process is capable)

where $C_{PL}(0)$ and $C_{PU}(0)$ are the recommended minimum values for these CIs in the presence of lower and upper one-sided tolerance specifications respectively (see Montgomery (2001)). They state that the critical values are identical for testing C_{PL} and C_{PU} and they provide accurate power values for various choices of C_{PL} or C_{PU} and n . For the case of two-sided tolerances, they recommend using the one-sided critical values as conservative critical values for testing

$$\begin{array}{ll} H_0 : C_{Pk} \leq C_{Pk}(0) & \text{(the process is not capable)} \\ \text{against} & \\ H_a : C_{Pk} > C_{Pk}(0) & \text{(the process is capable)} \end{array}$$

3. Distributions of Estimated Capability Indices by Conditioning

3.1 Estimators of C_{PL} and C_{PU}

For simplicity we first consider the one-sided indices C_{PL} and C_{PU} . Let $X_1, X_2, \dots, X_n$ be a random sample from the process under consideration when the process is in statistical control. The in-control process is assumed to be distributed as a normal $N(\mu; \sigma^2)$ with both μ and σ^2 unknown. This sample is considered to be the *reference sample* from which the unknown mean and variance are estimated using the sample mean $\bar{X}$ and the sample variance S^2 , respectively. The natural estimator of C_{PL} is $\hat{C}_{PL} = \frac{\bar{X} - L}{3S}$ and clearly the distribution of $\hat{C}_{PL}$ depends on the distributions of $\bar{X}$ and S^2 , which are normal and proportional to a χ^2 -distribution respectively. Our approach of deriving the cdf or the pdf of $\hat{C}_{PL}$, involves two steps:

1. Find the conditional distribution of the estimator $\hat{C}_{PL}$, given $S = s$;
2. Derive the unconditional distribution of $\hat{C}_{PL}$ by averaging over the marginal distribution of S .

This will be referred to as the *conditioning approach*, which takes advantage of the fact that for a random sample from the normal distribution, the sample mean and variance are statistically independent and their probability distributions are well known.

Consider the conditional cumulative distribution function of $\hat{C}_{PL}$, given $S = s$. Using normality of $\bar{X}$, we can write

$$F_L(t|S=s) \equiv P\left(\hat{C}_{PL} \leq t|s\right) = \Phi\left(\frac{\sqrt{n}(L-\mu)}{\sigma} + 3\sqrt{n}\frac{s}{\sigma}t\right)$$

where Φ denotes the standard normal cdf.

The unconditional cdf of $\hat{C}_{PL}$ can be obtained by averaging the conditional cdf over the distribution of S . However, we first re-write the conditional cdf of $\hat{C}_{PL}$ as

$$\Phi\left(\frac{\sqrt{n}(L-\mu)}{\sigma} + 3\sqrt{\frac{n}{n-1}}\sqrt{\frac{(n-1)s^2}{\sigma^2}}t\right)$$

and thus

$$F_L(t|Y=y) = \Phi\left(\frac{\sqrt{n}(L-\mu)}{\sigma} + 3\sqrt{\frac{n}{n-1}}\sqrt{yt}\right), \quad (1)$$

where $y = \frac{(n-1)s^2}{\sigma^2}$. The unconditional cdf of $\hat{C}_{PL}$ is then easily obtained from (1) by averaging (integrating) over the distribution of $Y = \frac{(n-1)S^2}{\sigma^2}$, which is of course the $\chi^2(n-1)$ distribution. Hence the cdf is given by

$$F_L(t) = \int_0^\infty \Phi\left(\frac{\sqrt{n}(L-\mu)}{\sigma} + 3\sqrt{\frac{n}{n-1}}\sqrt{yt}\right) f_{\chi^2(n-1)}(y) dy \quad (2)$$

Differentiating (1) with respect to t and averaging the result over the distribution of Y , we obtain the unconditional pdf of $\hat{C}_{PL}$

$$f_L(t) = \int_0^\infty \phi\left(\frac{\sqrt{n}(L-\mu)}{\sigma} + 3\sqrt{\frac{n}{n-1}}\sqrt{yt}\right) 3\sqrt{\frac{n}{n-1}}\sqrt{y} f_{\chi^2(n-1)}(y) dy \quad (3)$$

where ϕ denotes the pdf of the standard normal distribution and $f_{\chi^2(n-1)}$ denotes the pdf of the $\chi^2(n-1)$ distribution.

Remark. Since $\hat{C}_{PL}$ is, conditionally on $Y = y$, a linear combination in $\bar{X}$, a normal variate, one can easily show that $\hat{C}_{PL}$, given $Y = y$, is $N\left(C_{PL}\sqrt{\frac{n-1}{y}}; \frac{n-1}{9ny}\right)$ distributed, a fact that also follows directly from (1). The expected value of the unconditional distribution of $\hat{C}_{PL}$ then follows immediately from the moments of Y and is given by $E(\hat{C}_{PL}) = \sqrt{n-1} \frac{\Gamma\left(\frac{n-2}{2}\right)}{\sqrt{2}\Gamma\left(\frac{n-1}{2}\right)} C_{PL}$. This expression clearly illustrates that $\hat{C}_{PL}$ is not an unbiased estimator of C_{PL} .

The expressions for the cdf and the pdf, given in (2) and (3), are more direct to deal with when making statistical inference about the index C_{PL} , compared to those presented by Chow and Owen (1989) in terms of the non-central t -distribution, as was done in Chow, Owen and Borrego (1990).

Further, note that from the definition of C_{PL} the cdf in (2) and the pdf in (3) can be rewritten as

$$F_L(t) = \int_0^\infty \Phi\left\{3\sqrt{n}\left(-C_{PL} + \sqrt{\frac{y}{n-1}}t\right)\right\} f_{\chi^2(n-1)}(y) dy \quad (4)$$

$$f_L(t) = 3\sqrt{\frac{n}{n-1}} \int_0^\infty \phi\left\{3\sqrt{n}\left(-C_{PL} + \sqrt{\frac{y}{n-1}}t\right)\right\} \cdot \sqrt{y} f_{\chi^2(n-1)}(y) dy, \quad -\infty < t < \infty, \quad C_{PL} > 0 \quad (5)$$

respectively. The expressions in (4) and (5) clearly display the dependence of the distribution on the parameter C_{PL} .

Similarly, the cdf and the pdf of $\hat{C}_{PU}$ can be found and expressed as

$$\begin{aligned} F_U(t) &= 1 - \int_0^\infty \Phi \left\{ 3\sqrt{n} \left(C_{PU} - \sqrt{\frac{y}{n-1}} t \right) \right\} f_{\chi^2(n-1)}(y) dy \\ &= \int_0^\infty \Phi \left\{ 3\sqrt{n} \left(-C_{PU} + \sqrt{\frac{y}{n-1}} t \right) \right\} f_{\chi^2(n-1)}(y) dy \end{aligned} \quad (6)$$

$$\begin{aligned} f_U(t) &= 3\sqrt{\frac{n}{n-1}} \int_0^\infty \phi \left\{ 3\sqrt{n} \left(C_{PU} - \sqrt{\frac{y}{n-1}} t \right) \right\} \\ &\quad \cdot \sqrt{y} f_{\chi^2(n-1)}(y) dy \\ &= 3\sqrt{\frac{n}{n-1}} \int_0^\infty \phi \left\{ 3\sqrt{n} \left(-C_{PU} + \sqrt{\frac{y}{n-1}} t \right) \right\} \\ &\quad \cdot \sqrt{y} f_{\chi^2(n-1)}(y) dy, \quad -\infty < t < \infty, \quad C_{PU} > 0 \end{aligned} \quad (7)$$

Remark. Comparing, for example, (5) and (7), it is seen that the distributional forms of the two one-sided sample capability indices, $\hat{C}_{PL}$ and $\hat{C}_{PU}$, are identical. We submit that these derivations are more straightforward and provide more insight than those obtained in terms of a non-central t -distribution.

3.2 Estimator of C_{Pk}

Next we consider the more complicated case of the two-sided index C_{Pk} which is a function (minimum) of the two one-sided indices studied above. According to the literature, the C_{Pk} index is probably the most popular capability index

for measuring process capability. The inference problem and the associated distribution theory for the estimator $\hat{C}_{Pk}$ have presented interesting challenges and a number of approaches have been considered. Chow and Owen (1989) obtain the distribution in terms of the bi-variate non-central t -distribution and this theory has been the basis of several subsequent articles in the literature. We derive the distribution using the conditioning approach which allows us to express the distribution under a number of parameterizations used by various authors in the literature. This, in turn, helps our understanding of how the distribution depends on the unknown parameters.

As in the previous section, we first find the conditional cdf of the estimator $\hat{C}_{Pk}$, given $S = s$, as follows:

$$\begin{aligned}
 F_{\hat{C}_{Pk}}(t|S=s) &= P\left(\min\left(\hat{C}_{PL}; \hat{C}_{PU}\right) \leq t|s\right) \\
 &= P\left(\hat{C}_{PL} \leq t, \bar{X} < \frac{U+L}{2}\right) + P\left(\hat{C}_{PU} \leq t, \bar{X} > \frac{U+L}{2}\right), \quad \text{given } s \\
 &= \begin{cases} P(\bar{X} < L+3st) + P(\bar{X} > U-3st), & \text{given } s, \quad \text{for } t < \frac{U-L}{6s} = C_P \frac{\sigma}{s} \\ P\left(\bar{X} < \frac{U+L}{2}\right) + P\left(\bar{X} > \frac{U+L}{2}\right), & \text{given } s, \quad \text{for } t > C_P \frac{\sigma}{s} \end{cases} \\
 &= \begin{cases} F_L(t|S=s) + F_U(t|S=s), & \text{for } t < C_P \frac{\sigma}{s} \\ 1 & \text{for } t > C_P \frac{\sigma}{s} \end{cases},
 \end{aligned}$$

with $F_L(t|S=s)$ and $F_U(t|S=s)$ the conditional cdf of $\hat{C}_{PL}$ and $\hat{C}_{PU}$ respectively (see (1)). The restriction on the range is obtained from the facts that $L + 3st$ must be smaller than $U - 3st$ for the first probability statement and $C_P = \frac{(U-L)}{6\sigma}$.

Again, transforming: $y = \frac{(n-1)s^2}{\sigma^2}$ we can write the conditional cdf of $\hat{C}_{Pk}$ as

$$F_{\hat{C}_{Pk}}(t|Y=y) = \begin{cases} F_L(t|Y=y) + F_U(t|Y=y), & \text{for } t < C_P \sqrt{\frac{n-1}{y}} \\ 1 & , \text{ for } t > C_P \sqrt{\frac{n-1}{y}} \end{cases}$$

Hence, using indicator functions

$$I_{\{t|t < C_P \sqrt{\frac{n-1}{y}}\}} = \begin{cases} 1, & \text{if } t < C_P \sqrt{\frac{n-1}{y}} \\ 0, & \text{otherwise} \end{cases}$$

and

$$I_{\{t|t > C_P \sqrt{\frac{n-1}{y}}\}} = \begin{cases} 1, & \text{if } t > C_P \sqrt{\frac{n-1}{y}} \\ 0, & \text{otherwise} \end{cases}$$

and the fact that $1 - \Phi(x) = \Phi(-x)$ we have the conditional cdf of $\hat{C}_{Pk}$

$$\begin{aligned} F_{\hat{C}_{Pk}}(t|y) &= \left\{ \Phi \left(3\sqrt{\frac{n}{n-1}}\sqrt{y}t + \frac{\sqrt{n}(L-\mu)}{\sigma} \right) \right. \\ &\quad \left. + \Phi \left(3\sqrt{\frac{n}{n-1}}\sqrt{y}t - \frac{\sqrt{n}(U-\mu)}{\sigma} \right) \right\} \\ &\quad \cdot I_{\{t|t < C_P \sqrt{\frac{n-1}{y}}\}} + I_{\{t|t > C_P \sqrt{\frac{n-1}{y}}\}} \end{aligned} \quad (8)$$

We now provide a series of equivalent expressions for the conditional cdf of $\hat{C}_{Pk}$ implementing various (re-)parameterizations used in the literature.

$$F_{\hat{C}_{Pk}}(t|y) = \left\{ \begin{array}{l} \left\{ \Phi \left(3\sqrt{y}t + \frac{\sqrt{n}(L - \mu)}{\sigma} \right) \right. \\ \quad \left. + \Phi \left(3\sqrt{\frac{n}{n-1}}\sqrt{y}t - \frac{\sqrt{n}(U - \mu)}{\sigma} \right) \right\} \\ \quad \cdot I_{\left\{ t|t < C_P\sqrt{\frac{n-1}{y}} \right\}} + I_{\left\{ t|t > C_P\sqrt{\frac{n-1}{y}} \right\}}; \quad (8a) \\ \left\{ \Phi \left(3\sqrt{\frac{n}{n-1}}\sqrt{y}t - 3\sqrt{n}C_{PL} \right) \right. \\ \quad \left. + \Phi \left(3\sqrt{\frac{n}{n-1}}\sqrt{y}t - 3\sqrt{n}C_{PU} \right) \right\} \\ \quad \cdot I_{\left\{ t|t < C_P\sqrt{\frac{n-1}{y}} \right\}} + I_{\left\{ t|t > C_P\sqrt{\frac{n-1}{y}} \right\}}; \quad (8b) \\ \left\{ \Phi \left(3\sqrt{n} \left(C_{Pk} - 2C_P + \sqrt{\frac{y}{n-1}}t \right) \right) \right. \\ \quad \left. + \Phi \left(3\sqrt{n} \left(-C_{Pk} + \sqrt{\frac{y}{n-1}}t \right) \right) \right\} \\ \quad \cdot I_{\left\{ t|t < C_P\sqrt{\frac{n-1}{y}} \right\}} + I_{\left\{ t|t > C_P\sqrt{\frac{n-1}{y}} \right\}}; \quad (8c) \\ \left\{ \Phi \left(\sqrt{n}(-\delta - |\gamma|) + 3\sqrt{\frac{n}{n-1}}\sqrt{y}t \right) \right. \\ \quad \left. + \Phi \left(\sqrt{n}(-\delta + |\gamma|) + 3\sqrt{\frac{n}{n-1}}\sqrt{y}t \right) \right\} \\ \quad \cdot I_{\left\{ t|t < C_P\sqrt{\frac{n-1}{y}} \right\}} + I_{\left\{ t|t > C_P\sqrt{\frac{n-1}{y}} \right\}} \quad (8d) \end{array} \right.$$

where $\delta = 3C_P$ and $|\gamma| = \left| \mu - \frac{U+L}{2} \right| / \sigma$.

It is important to note that each of the above expressions depends on 2 parameters (e.g. μ and σ , C_P and C_{P_k} or C_{PL} and C_{PU}). Recall that $C_{P_k} \leq C_P$, $C_{P_k} \leq C_{PL}$ and $C_{P_k} \leq C_{PU}$.

Some simplifications result when $C_{PL} = C_{PU}$. In this case $C_{PL} = C_{PU} = C_P = C_{P_k} = \theta$ say, and we can write

$$F_{\hat{C}_{P_k}}(t|y) = 2 \left\{ \Phi \left(-3\sqrt{n}\theta + 3\sqrt{\frac{ny}{n-1}}t \right) \right\} I_{\left\{ t|t < \theta\sqrt{\frac{n-1}{y}} \right\}} + I_{\left\{ t|t > \theta\sqrt{\frac{n-1}{y}} \right\}}$$

Note that in (8c) we use the fact that either $C_{P_k} = C_{PL}$ or $C_{P_k} = C_{PU}$ and in both cases the cdf of $\hat{C}_{P_k}$ reduces to

$$\begin{aligned} F_{\hat{C}_{P_k}}(t|y) &= \left\{ \Phi \left(3\sqrt{n} \left[C_{P_k} - 2C_P + \sqrt{\frac{y}{n-1}}t \right] \right) \right. \\ &\quad \left. + 1 - \Phi \left(3\sqrt{n} \left[C_{P_k} - \sqrt{\frac{y}{n-1}}t \right] \right) \right\} \\ &\quad \cdot I_{\left\{ t|t < C_P\sqrt{\frac{n-1}{y}} \right\}} + I_{\left\{ t|t > C_P\sqrt{\frac{n-1}{y}} \right\}} \end{aligned}$$

The unconditional cdf of $\hat{C}_{P_k}$ is then found by integrating (averaging) the conditional cdf over the distribution of Y :

$$F_{\hat{C}_{P_k}}(t) = \int_0^\infty F_{\hat{C}_{P_k}}(t|Y=y) f_{\chi^2(n-1)}(y) dy \quad (9)$$

The unconditional pdf of $\hat{C}_{P_k}$ is found by integrating the derivative of the conditional cdf over the distribution of Y or the derivative of the unconditional cdf of $\hat{C}_{P_k}$ given in (9). For example, corresponding to (8c) we have

$$\begin{aligned}
f_{\hat{C}_{Pk}}(t) &= 3\sqrt{\frac{n}{n-1}} \int_0^{\frac{(n-1)}{t^2} C_P^2} \left\{ \phi \left(3\sqrt{n} \left[C_{Pk} - 2C_P + \sqrt{\frac{y}{n-1}} t \right] \right) \right. \\
&\quad \left. + \phi \left(3\sqrt{n} \left[C_{Pk} - \sqrt{\frac{y}{n-1}} t \right] \right) \right\} \sqrt{y} f_{\chi^2(n-1)}(y) dy \\
&\quad -\infty < t < \infty, \quad 0 < C_{Pk} \leq C_P
\end{aligned} \tag{10}$$

Remarks.

1. Note that one can also express the cdf of $\hat{C}_{Pk}$ in terms of either the cdf of the univariate non-central t distribution or the cdf of the bivariate non-central t -distribution. We provide these expressions below for completeness.

$$F_{\hat{C}_{Pk}}(t) = \begin{cases} F_{T_{n-1}}(3\sqrt{nt}, \delta_1 = 3\sqrt{n}C_{PL}) \\ \quad + 1 - F_{T_{n-1}}(-3\sqrt{nt}, \delta_2 = -3\sqrt{n}C_{PU}) \\ \quad 1 - P(T_{n-1}(\delta_1 = -3\sqrt{n}C_{PU}) \\ \quad \leq -3\sqrt{nt}, T_{n-1}(\delta_2 = 3\sqrt{n}C_{PL}) \geq 3\sqrt{nt}) \end{cases} \tag{11}$$

The second of these expressions was provided in Chow and Owen (1989). Obviously these expressions are difficult to evaluate without additional means to deal with the univariate or bivariate non-central t -distributions and various approximations have been considered in the literature. To the contrary, our expressions for the distribution given in (8a) through (8d) are more direct and arguably more convenient for evaluation. We have used *Mathcad* and *Mathematica* numerical integration routines for the calculations in this paper. The new expressions also provide more insight into the properties of the relevant distributions (for example how the pdf or the cdf depends on the unknown parameters). It may be noted that these integrals can also be estimated by simulation from a normal distribution.

2. The method followed in this paper to find the distribution of a sample capability index is based on two steps, namely finding the conditional distribution of the sample capability index, given $Y = y$ (or $S = s$,

where $Y = \frac{(n-1)S^2}{\sigma^2}$ firstly, and secondly, averaging out over the marginal distribution of Y . One could instead firstly derive the conditional distribution of the sample capability index, given $\bar{X} = \bar{x}$ (or given $Z = z$, with $Z = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}$) and secondly, averaging out over the marginal distribution of Z . The resulting expression for the distribution of $\hat{C}_{Pk}$, for example, would be essentially the same as that obtained by Vännman (1997) in Theorem 3 for the special case of $u = 1$, $v = 0$ in her general class of capability indices.

3. Yet another way of deriving the conditional density function of $\hat{C}_{Pk}$, given $Y = y$, is to make use of the alternative expression $\hat{C}_{Pk} = \frac{U - L - 2|\bar{X} - M|}{6S}$, $M = \frac{U + L}{2}$ and using the fact that $\bar{X}$ is a normal variate.

4. Applications: Moments and Statistical Inference

Currently, the available cdfs and the pdfs of the above-mentioned sample CIs, with forms of varying complexity (the case of $\hat{C}_{Pk}$ being the most complex one) are being used in the literature and some tables and computer programs are available. See for example, Kotz and Lovelace (1998) or Kotz and Johnson (2002) for a review. From these results, the moments of the sample indices as well as exact and/or approximate confidence intervals and critical values for testing certain hypotheses can be derived under various assumptions (such as approximate normality of the sampled capability indices).

In our opinion and as we demonstrate below, the exact distributional results given above (Section 3) should make it much more straightforward to calculate these quantities such as moments, critical values and confidence limits. We provide some examples in the next section.

4.1 The moments of $\hat{C}_P$ and $\hat{C}_{Pk}$

In Table 2.3 of Kotz and Johnson (1993) the mean and standard deviation of $\hat{C}_P$ can be found. In their Table 2.4 the mean and standard deviation of $\hat{C}_{Pk}$ are given for the following parameter combinations:

- $|\gamma| = \left| \mu - \frac{U+L}{2} \right| / \sigma = \frac{3}{2} |C_{PU} - C_{PL}| = 3 |C_P - C_{Pk}| = 0 (0.5) 2$
- $\delta = \frac{U-L}{2\sigma} = 3C_P = 2 (1) 6$
- $n = 10 (5) 20, 55 \text{ and } 60$

As an illustration of the application of our distributional formulas, we recalculated the values of their Table 2.4 for a variety of parameter combinations as well as the coefficient of skewness (which was not reported in their table). In our calculations we made use of both *Mathcad* and *Mathematica*, which in our opinion are suitable packages for this purpose, since we needed a simple and accurate method for numerical integration. Programming is easy and our values agreed with theirs, with a few exceptions where we differed in the 3rd decimal place.

Using our expressions for the density functions, we can, for example, easily calculate various moments of the distribution.

Example 1: Suppose $C_P = C_{PL} = C_{PU} = C_{Pk} = 2/3$, $n = 20$. For the $\hat{C}_{Pk}$ distribution we found the expected value, the standard deviation and the skewness coefficient to be 0.632554, 0.11913, and 0.795765, respectively. In Figure 1 we show the pdf of $\hat{C}_{Pk}$ for the parameter values mentioned above and compare it with a normal pdf with mean 0.632554 and standard deviation 0.11913 to visually assess the validity of the assumption that $\hat{C}_{Pk}$ approximately follows a normal distribution.

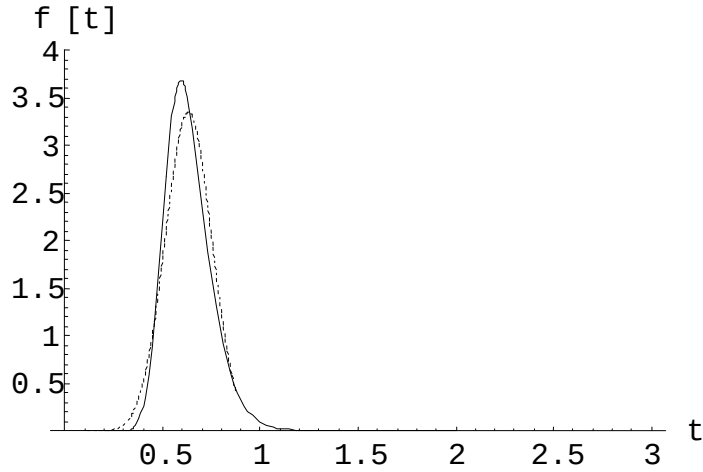


Figure 1. Probability density functions of $\hat{C}_{P_k}$ (solid line) and normal distribution (dashed line), with mean 0.632554 and standard deviation 0.11913

It can be seen from Figure 1 that the distribution of $\hat{C}_{P_k}$ is different from that of the normal, particularly in the tails. Also, since the skewness coefficients in many cases (various sample sizes and parameter combinations) can be seen to be not close to 0, we conclude that the assumption of normality for the distribution of $\hat{C}_{P_k}$ is not a safe one for small to moderate sized samples. Thus considering a normal approximation (by several authors, including Bissell (1990), Zhang et al. (1990)) for the distribution of $\hat{C}_{P_k}$ to make statistical inference on C_{P_k} (testing as well as confidence intervals) may be problematic, especially for smaller sample sizes. Therefore, we recommend exact (small sample) inference based on the exact distribution of the estimators.

4.2 Inference for C_{PL} and C_{PU}

The distribution theory developed above (Section 3) can be used to set up suitable statistical inference procedures. For example, we consider confidence intervals that are very useful tools in making statistical inference. One-sided confidence intervals are defined in terms of a single confidence limit (also called a confidence bound) and typically these may be more useful in the context of process capability.

We now consider accurate (small sample) confidence limits for the one-sided indices.

4.2.1 Confidence limits

Consider the index C_{PL} first. From (4), it can easily be seen that the cdf of $\hat{C}_{PL}$ is a decreasing function of C_{PL} . Thus, a lower 95% confidence limit for C_{PL} can be found from an observed value $\hat{C}_{PL}$ by solving (using *Mathcad* or *Mathematica*) $F_L(\hat{C}_{PL}) = 0.95$ for C_{PL} . As an example, suppose that a random sample of size 60 was available from a $N(0, 0.6^2)$ distribution and the observed value of $\hat{C}_{PL} = 1.56874$ was found using specification limits $L = -3$ and $U = 3$. Solving the equation $F_L(1.56874) = 0.95$ for C_{PL} , we find the 95% lower confidence limit 1.31825 for C_{PL} . Note that expression (4) makes it much more straightforward to solve for C_{PL} compared to when the relationship is expressed in terms of the non-central t -distribution.

Using this lower confidence limit, we also see, for example, that the null hypothesis $H_0 : C_{PL} \leq C_{PL}(0) = 1.25$ can be rejected in favour of $H_a : C_{PL} > C_{PL}(0)$. Therefore, we can conclude that the process is capable based on the given sample.

Similarly, a lower 95% confidence limit for C_{PU} can be found from an observed value $\hat{C}_{PU}$ by solving $F_U(\hat{C}_{PU}) = 0.95$ for C_{PU} . For the same sample of size 60 and the same specification limits mentioned above, an

observed value $\hat{C}_{PU} = 1.71687$ was found. Solving $F_U(1.71687) = 0.95$ for C_{PU} we get the 95% lower confidence limit 1.44462 for C_{PU} .

Chou, Owen and Borrego (1990) provide tables for the 95% lower confidence limit for C_{PL} (and C_{PU}) for sample sizes $n = 10, 20, 30, 40, 50, 75, 100, 125, 150, 200, 250, 300, 350$ and 400 and observed values of $\hat{C}_{PL} = 0.7(0.1)3.0$. We could easily replicate the values in their Table 4. Process control practitioners typically need values like these to get $100\gamma\%$ lower confidence interval for C_{PL} (or C_{PU}). However, when the confidence level γ differs from 0.95 and/or the sample size n or the observed $\hat{C}_{PL}$ (or $\hat{C}_{PU}$) values differ from their tabulated values, an approximation will be in practice. Instead, using our approach makes finding the confidence limit much easier and more accurate for the practitioner. One can simply plug in the values of γ , n , $\hat{C}_{PL}$ (or $\hat{C}_{PU}$) and solve for C_{PL} (or C_{PU}) exactly, as illustrated above. The solution can be obtained by either numerical methods (as we did) or by simulation, if one should prefer that.

4.2.2 Hypothesis testing, Power and Sample size

Chou and Polansky (1993) give critical values for testing the null hypothesis that the process is not capable against the alternative that it is capable (see Section 2 above). Using these critical values, given in their Table I, implies that by rejecting H_0 one may conclude (on the relevant level of significance) that the sample estimate of the relevant capability index is significantly higher than the hypothesized value of its parametric counterpart. The critical values in their Table I can be obtained from the exact null distribution of $\hat{C}_{PL}$ and $\hat{C}_{PU}$. Using our conditioning approach, we recalculated some critical values and power values for the case of $\hat{C}_{PL}$ (and $\hat{C}_{PU}$) and could confirm their correctness. Again, our approach is more direct and straightforward to implement in practice.

One important advantage of our approach, concerns sample size determination. The numerical solution approach allows us to solve directly for n (sample size) when the required power of the test of $H_0 : C_{PL} \leq C_{PL}(0)$ is specified. This is obtained by jointly solving

$$1 - \int_0^{\infty} \left\{ \Phi \left(3\sqrt{n} \left(-C_{PL}(0) + \sqrt{\frac{y}{n-1}} t(\alpha, n, C_{PL}(0)) \right) \right) \right\} f_{x^2(n-1)}(y) dy = \alpha \quad (12)$$

and

$$1 - \int_0^{\infty} \left\{ \Phi \left(3\sqrt{n} \left(-C_{PL}(1) + \sqrt{\frac{y}{n-1}} t(\alpha, n, C_{PL}(0)) \right) \right) \right\} f_{x^2(n-1)}(y) dy = \pi \quad (13)$$

for the critical value $t(\alpha, n, C_{PL}(0))$ and the sample size n , for a given α (the desired significance level), π (the required power) and $C_{PL}(1)$ (the value of the CI to be detected under the alternative $C_{PL}(1) > C_{PL}(0)$). We illustrate the sample size determination for a few examples with the results shown in Table 1 below.

Table 1. Sample sizes for $C_{PL}(0) = 1.25$, $C_{PL}(1) = 2.0$, $\alpha = 0.05$ and various values for power

Required power	Critical value	Sample size, n
0.6	1.950	12
0.8	1.768	18
0.9	1.683	24

Thus, if the practitioner needs to find the required sample size to detect a C_{PL} value of 2.0, using a hypothesised value of 1.25 at a level of significance 0.05 and a power of 0.8, he needs a sample of 18 and the resulting critical value for the test will be 1.768. We strongly believe that easily calculating a required

sample size should be most useful to the practitioner, planning improvements to the process under study.

4.3 Joint inference for C_P and C_{Pk}

The indices C_P and C_{Pk} are among the most popular CIs in practice. Kotz and Lovelace (1998, page 68) state that these indices "...provide the extra advantage of including a direct comparison to the specification range, as well as providing a unitless measure of capability which makes processes throughout the organization comparable on a single scale." They go on to say that "They are by far the most popular of all the indices and it is expected that they will continue to be so despite the introduction of newer, and in some ways better, indices over the last few years. Multi-million dollar decisions are made every day based on the calculated values of these indices, making it extremely critical for them to be estimated properly, with the appropriate qualifications." As we have noted earlier, attempts have been made, under various assumptions, to make statistical inference on the indices C_{Pk} and C_P individually. However, more fundamentally, it has been noted that neither of these indices alone is sufficient to reflect the property of capability of an in-control process (see, for example, Chapter 8, Farnum (1994) and Chapter 7, Montgomery (2001)). In fact, again quoting Kotz and Lovelace (1998), "It is evident... that C_{Pk} and C_P , when used together, give a very good indication of how well a process is now performing and how well a process could perform if it were centered on target." Thus one should consider these indices together and not individually, since while C_P depends only on the process standard deviation and ignores the centre or the mean, the effects of the mean and the standard deviation are confounded in C_{Pk} (as in C_{PL} and C_{PU}).

The following simple example makes it clear: Consider two processes for which $U = 3$, $L = -3$ and with a target value of 0. Suppose that the

estimates $\bar{x} = 1.5$, $s = 0.5$ were observed for Process 1 and that $\bar{x} = 0.6$ and $s = 0.8$ were observed for Process 2. Both of these pairs yield a $\hat{C}_{Pk}$ value of 1.0, which, if judged on the value of $\hat{C}_{Pk}$ alone, would mean a borderline capable process in each case. However, the $\hat{C}_P$ values are different, namely 2 and 1.25, respectively, and these point to Process 1 as the one with higher potential capability. The extra information from $\hat{C}_P$, (namely that $\hat{C}_P = 2$ and $\hat{C}_P = 1.25$), sheds more light on the process's capability and allows one to make a choice. Kotz and Lovelace (1998, page 64) state "Using either C_P or C_{Pk} provides only part of the story, since both indices have their individual weaknesses. However, by examining the relationship between them and using these indices as a pair, a tremendous amount of information can be gleaned about the process without the worrisome confounding of μ and σ in the C_{Pk} case." They suggest using both $\hat{C}_P$ and $\hat{C}_{Pk}$ in the analysis of capability and show that the proportion of non-conforming items, p , can be expressed in terms of both C_P and C_{Pk} : $p = \Phi[-3(2C_P - C_{Pk})] + \Phi[-3C_{Pk}]$. Our expression in (8c) explicitly shows that the distribution of $\hat{C}_{Pk}$ depends on both C_{Pk} and C_P . We therefore propose that in process capability analysis, statistical inference (estimation or hypothesis testing) should be aimed at the vector of indices (C_P, C_{Pk}) together and not on either one of the two indices.

In the next section we develop exact inference procedures for the vector parameter (C_P, C_{Pk}) .

4.3.1 Hypothesis Testing

We start with hypothesis testing. As explained above, in order to evaluate a process's capability, inference should be aimed at estimating or testing C_P and C_{Pk} jointly. In general, large values of both C_P and C_{Pk} imply capability. Since we are dealing with a bivariate parameter $(C_P; C_{Pk})$ and its sample estimator $(\hat{C}_P; \hat{C}_{Pk})$, the bivariate region of possible (C_P, C_{Pk}) values

needs to be investigated. To this end, consider a case where, for example, both C_P and C_{Pk} must be greater than 1.33 for the process to be capable. Now, consider Figure 2 in which the area marked 1a depicts the region where both C_P and C_{Pk} are larger than 1.33, implying a capable process in both parameters. If we take the point (1.33, 1.33) as the origin for the 4 numbered quadrants, points in quadrants 1b, 2 and 3a are impossible (since $C_P \leq C_{Pk}$), quadrant 3b reflects the region of incapability and quadrant 4 implies capability in only the C_P -sense. Note that there are only 3 regions relevant, namely 1a, 4 and 3b, with 3b representing incapability, 1a representing full capability and 4 a partly capable process. It is clear that only quadrant 1a represents points where the process is capable in both senses, as one would typically require.

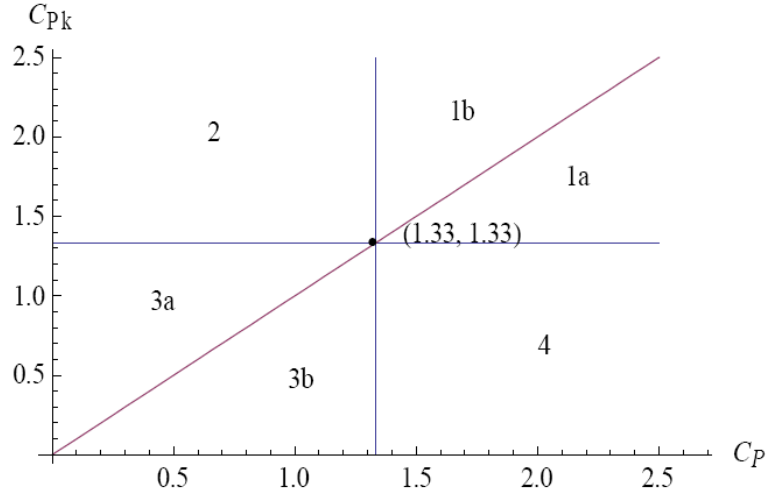


Figure 2. Region of possible $(C_P; C_{Pk})$ values

Now, consider in general testing the null hypothesis

$$H_0 : (C_P, C_{Pk}) \leq \{C_P(0), C_{Pk}(0)\},$$

where $\{C_P(0), C_{Pk}(0)\}$ is some specified vector of values, such as (1.33, 1.33), against the alternative hypothesis

$$H_1 : H_0 \text{ not true.}$$

From the justification above, accepting H_0 (concluding that (C_P, C_{Pk}) lies in region 3b of Figure 2) would mean that the process is not capable. On the other hand, rejecting H_0 in favour of H_1 means that the process is capable in at least the C_P -sense ((C_P, C_{Pk}) lies in region 1a or 4 of Figure 2).

To conduct hypothesis testing, based on an observed sample from the in-control process, we propose to calculate the p -value,

$$P \left\{ \left(\hat{C}_P, \hat{C}_{Pk} \right) \geq \left(\hat{C}_P(\text{obs}), \hat{C}_{Pk}(\text{obs}) \right) \mid (C_P, C_{Pk}) \in H_0 \right\},$$

which is the probability of “exceedance” of the observed value $\left(\hat{C}_P(\text{obs}), \hat{C}_{Pk}(\text{obs}) \right)$ under H_0 . When this exceedance probability is small, it implies that the process is capable in at least the C_P -sense, since the observed vector lies in an unlikely region under H_0 . In other words, when the p -value is smaller than a specified significance level α , H_0 is rejected in favour of H_1 . Figure 3 illustrates one such example, where $\left(\hat{C}_P(\text{obs}), \hat{C}_{Pk}(\text{obs}) \right) = (1.84, 1.18)$. Similar to Figure 2, we divided the plane into 4 quadrants, numbered 5 to 8 (regions 5b, 6a and 7a are not relevant, since $\hat{C}_P \leq \hat{C}_{Pk}$). Note that the events $\left(\hat{C}_P, \hat{C}_{Pk} \right) \geq \left(\hat{C}_P(\text{obs}), \hat{C}_{Pk}(\text{obs}) \right)$ (region 5a) and $\left(\hat{C}_P, \hat{C}_{Pk} \right) < \left(\hat{C}_P(\text{obs}), \hat{C}_{Pk}(\text{obs}) \right)$ (region 7b) are not complementary. However, also note that one would not associate high overall capability with the complement of $\left(\hat{C}_P, \hat{C}_{Pk} \right) < \left(\hat{C}_P(\text{obs}), \hat{C}_{Pk}(\text{obs}) \right)$, (i.e. regions 5a, 6b and 8) but rather with the event $\left(\hat{C}_P, \hat{C}_{Pk} \right) \geq \left(\hat{C}_P(\text{obs}), \hat{C}_{Pk}(\text{obs}) \right)$

(region 5a), since the regions 6b and 8 would include $(\hat{C}_P, \hat{C}_{Pk})$ values coming from a process which is possibly not capable in both senses.

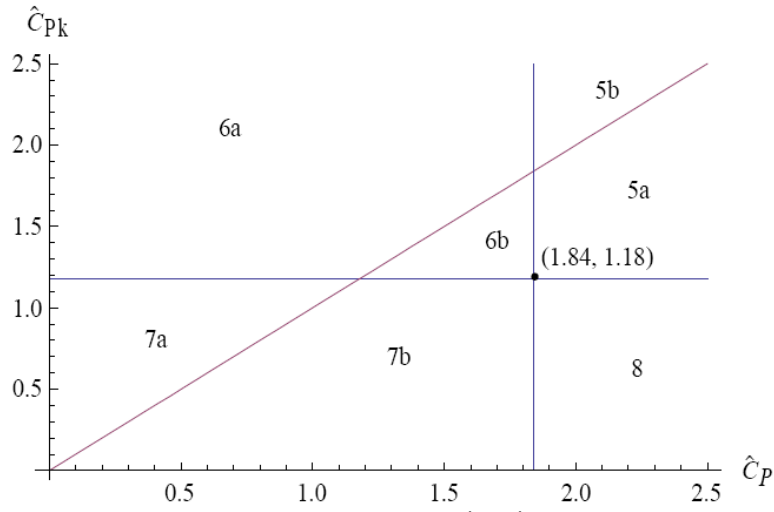


Figure 3. Region of possible $\hat{C}_P$; $\hat{C}_{Pk}$ values

Therefore, we would reject H_0 if region 5a has a probability less than the level of significance α . Finally, also note that the null hypothesis is composite and therefore this testing procedure is valid if the p -value is less than the significance level for all $(C_P, C_{Pk}) \in H_0$. This will be true since $P\left\{(\hat{C}_P, \hat{C}_{Pk}) \geq (r, t)\right\}$ is non-decreasing in both C_P and C_{Pk} (see comment at the end of Section 4.3 2).

In order to implement this procedure, we first need to find the joint probability distribution of $(\hat{C}_P, \hat{C}_{Pk})$. This is obtained as follows.

4.3.2 Joint distribution of $\hat{C}_P$ and $\hat{C}_{Pk}$

Since $C_{Pk} \leq C_P$ in general, we let $t \leq r$ and consider

$$\begin{aligned}
& P\left(\widehat{C}_P \geq r, \widehat{C}_{Pk} \geq t\right) \\
&= P\left(\frac{U-L}{6S} \geq r, \widehat{C}_{Pk} \geq t\right) \\
&= P\left(Y \leq \frac{(n-1)C_P^2}{r^2}, \widehat{C}_{Pk} \geq t\right), \text{ where } Y \text{ has a } \chi^2(n-1) \\
&\text{distribution.}
\end{aligned}$$

If we condition on S or Y as before, the two inequalities in the previous expression reduce to one inequality in the random variable $\widehat{C}_{Pk}$ and one inequality on the conditioned value y , as follows:

$$\begin{aligned}
& P\left(\widehat{C}_P \geq r, \widehat{C}_{Pk} \geq t|y\right), t \leq r \\
&= P\left(\widehat{C}_{Pk} \geq t|y \leq \frac{(n-1)C_P^2}{r^2}\right) \\
&= \left\{1 - \Phi\left(3\sqrt{n}\left(C_{Pk} - 2C_P + \sqrt{\frac{y}{n-1}}t\right)\right)\right. \\
&\quad \left.- \Phi\left(3\sqrt{n}\left(-C_{Pk} + \sqrt{\frac{y}{n-1}}t\right)\right)\right\} I_{\left\{y \leq \frac{(n-1)C_P^2}{r^2}\right\}} \\
&= \left\{\Phi\left(3\sqrt{n}\left(C_{Pk} - \sqrt{\frac{y}{n-1}}t\right)\right)\right. \\
&\quad \left.- \Phi\left(3\sqrt{n}\left(C_{Pk} - 2C_P + \sqrt{\frac{y}{(n-1)}}t\right)\right)\right\} I_{\left\{y \leq \frac{(n-1)C_P^2}{r^2}\right\}} \quad (14)
\end{aligned}$$

using 8(c).

Therefore, integrating over the marginal density of Y , the unconditional value of $P\left(\widehat{C}_P \geq r, \widehat{C}_{Pk} \geq t\right)$ is obtained:

$$\begin{aligned} & P\left(\widehat{C}_P \geq r, \widehat{C}_{Pk} \geq t\right) \\ &= \int_0^{\frac{(n-1)C_P^2}{r^2}} \left\{ \Phi\left(3\sqrt{n}\left(C_{Pk} - \sqrt{\frac{y}{n-1}}t\right)\right) \right. \\ & \quad \left. - \Phi\left(3\sqrt{n}\left(C_{Pk} - 2C_P + \sqrt{\frac{y}{n-1}}t\right)\right) \right\} f_{\chi^2(n-1)}(y) dy \quad (15) \end{aligned}$$

Using this last expression, the exact p -value for the observed values $\left(\widehat{C}_P(obs), \widehat{C}_{Pk}(obs)\right)$ can be found by substituting these observed values for r and t , respectively, into (15) along with the hypothesized $C_P = C_P(0)$ and $C_{Pk} = C_{Pk}(0)$. It can easily be seen from (15) that $P\left\{\left(\widehat{C}_P, \widehat{C}_{Pk}\right) \geq (r, t)\right\}$ is non-decreasing in both C_P and C_{Pk} .

4.3.3 Confidence region estimation

Using the relationship between the acceptance region of a test of hypothesis and a confidence region, our approach is as follows: Consider a random sample for which $\left(\widehat{C}_P(obs), \widehat{C}_{Pk}(obs)\right)$ is found. The $100(1 - \alpha)\%$ confidence region for (C_P, C_{Pk}) is defined by the set of points $(C_P(0), C_{Pk}(0))$ in the (C_P, C_{Pk}) plane below the line $C_P = C_{Pk}$, which cannot be rejected on a $100\alpha\%$ level of significance in favour of the alternative to $H_0 : (C_P, C_{Pk}) \leq C_P(0), C_{Pk}(0)$, using the p -value defined above in Section 4.3.1.

The rationale behind this approach is as follows: If the probability of exceedance of the observed value $\left(\widehat{C}_P(obs), \widehat{C}_{Pk}(obs)\right)$ is greater than α for a given value of the parameter vector, say $(C_P(1), C_{Pk}(1))$, in the direction stipulated by the alternative hypothesis, then $(C_P(1), C_{Pk}(1))$ can be seen to be an acceptable value, that is, it lies in the confidence region for the parameter vector. In other words, if $C_{Pk}(1) \leq C_P(1)$

and $P \left\{ \left(\hat{C}_P, \hat{C}_{P_k} \right) \geq \left(\hat{C}_P (obs), \hat{C}_{P_k} (obs) \right) \mid (C_P (1), C_{P_k} (1)) \right\} \geq \alpha$,
 $(C_P (1), C_{P_k} (1))$, is a plausible value of the parameter vector.

Thus, in order to find a $100(1 - \alpha)\%$ confidence region for (C_P, C_{P_k}) , we seek the set of points (C_P, C_{P_k}) satisfying the following two inequalities:

$$\begin{cases} P \left\{ \left(\hat{C}_P, \hat{C}_{P_k} \right) \geq \left(\hat{C}_P (obs), \hat{C}_{P_k} (obs) \right) \mid (C_P, C_{P_k}) \right\} \geq \alpha \\ \text{and} \\ 0 < C_{P_k} \leq C_P \end{cases} \quad (16)$$

Each point (C_P, C_{P_k}) satisfying (16) lies in the $100(1 - \alpha)\%$ confidence region. Note that the confidence region is a “lower” type (closed on the left and open on the right-hand side) which is really what is important in the process capability context.

4.4 Example

4.4.1 Test of Hypothesis

A random sample of size 60 was simulated from a $N(1, 0.6^2)$ distribution and $\left(\hat{C}_P (obs), \hat{C}_{P_k} (obs) \right) = (1.83649, 1.18352)$ was calculated, with specification limits $L = -3$ and $U = 3$ for a target value of 0. We take, for example, a value of 1.33 as an indication of a capable process, so that the null hypothesis $H_0 : (C_P, C_{P_k}) \leq (1.33; 1.33)$ (the process is not capable – region 3b in Figure 2) is tested against the alternative $H_1 : H_0$ is not true (the process is capable in at least the C_P -sense – regions 1a and 4 in Figure 2). The p -value for the observed $\left(\hat{C}_P (obs), \hat{C}_{P_k} (obs) \right)$ vector was found from (15) to be 0.001 (region 5a in Figure 3), so that H_0 is rejected at the 5% level of significance in favour of H_1 . This means that the process is capable, in at least the C_P -sense (see further discussions below).

4.4.2 Confidence region

Next, we calculate the 95% confidence region for (C_{Pk}, C_P) . Since $C_{Pk} \leq C_P$ must be satisfied, a natural starting point in this development is to substitute $(\hat{C}_P(obs), \hat{C}_{Pk}(obs))$ into (16) and solve for (C_P, C_{Pk}) from:

$$\begin{cases} P \left\{ \left(\hat{C}_P, \hat{C}_{Pk} \right) \geq \left(\hat{C}_P(obs), \hat{C}_{Pk}(obs) \right) \mid (C_P, C_{Pk}) \right\} = 0.05 \\ \text{and} \\ 0 < C_{Pk} = C_P \end{cases} \quad (17)$$

Using $(\hat{C}_P(obs), \hat{C}_{Pk}(obs)) = (1.83649, 1.18352)$ and applying numerical solution techniques (again with *Mathematica* and *Mathcad*), we obtain from (17): $(C_P, C_{Pk}) = (1.55573, 1.55573)$. The next step in the search process is to take any value above 1.55573 for C_P i.e., $C_P = C_P(1)$, with $C_P(1) > 1.55573$ and solve for C_{Pk} from the first condition in (17). By systematically repeating this last step we find the required C_{Pk} -values and use them to draw the (lower) borderline of the confidence region. A graphical representation of the 95% confidence region for (C_P, C_{Pk}) is displayed in Figure 4. It is seen that the point (1.33, 1.33) is not a member of the confidence region, but the point (1.8, 0.99) (for example) is included, which implies that the process is capable in the C_P -sense only. For the practitioner these findings mean that the process has potential (the variation in values of the process characteristic is small), but the process needs to be centred (the mean is not in the centre).

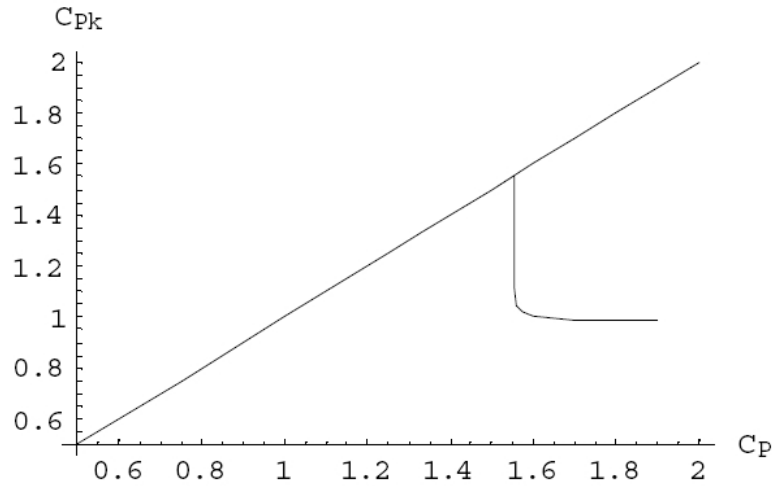


Figure 4. Confidence region for (C_P, C_{Pk})

5. Summary and concluding remarks

In this paper we use the method of conditioning to derive exact expressions for the cumulative distribution functions and the density functions of the four most commonly used sample capability indices. These expressions are easy to work with and more insightful when one wants to study properties of these sample capability indices as random variables. Using our approach, moments as well as the skewness coefficient of the estimator $\hat{C}_{Pk}$ can easily be calculated. The skewness figures as well as the graphs of the density functions cast doubt on the assumption of a normal approximation for the distribution of $\hat{C}_{Pk}$ for small to moderate size samples. This raises questions about some of the approximate inference methods available in the literature, and small sample accurate methods, based on our expressions derived here, should be preferred. In the context of hypothesis testing, critical values, power and

sample sizes for the one-sided indices C_{PU} and C_{PL} are also easily obtained via the conditioning approach. Confidence limits for the one-sided indices can also be obtained fairly easily.

It is emphasized that inference on the popular capability indices C_P or C_{Pk} , individually, may not be very meaningful in a process capability context. In fact, an inference on C_{Pk} alone is not possible since as is shown here, the sampling distribution of the estimator depends on both C_{Pk} and C_P . Hence these parameters need to be considered jointly when testing hypotheses or finding confidence intervals/regions. To this end, we develop hypothesis testing procedures and confidence regions for the vector (C_P, C_{Pk}) by first finding the joint pdf of the corresponding estimators. It does not seem meaningful to test the capability of a process based on one of the indices alone because two process parameters μ and σ are involved in each. If the process is found not capable, for example, it is not clear what needs to be done for improvement: Reduce variation, or shift the centre, or both. Similar arguments apply when making inference on the one-sided indices C_{PU} and C_{PL} (in spite of the fact that the density functions of $\hat{C}_{PU}$ and $\hat{C}_{PL}$ depend only on the relevant capability index parameter, as seen above). When looking at C_{PU} , for example, the same example as the one given in Section 4.3 in this context shows that one may obtain a process which is only marginally capable in the C_{PU} -sense, but when the additional information available in C_P is taken into account, it is seen that the process is potentially capable, so that the centre should be shifted to the right to increase C_{PU} and thus improve the process. Finally, note that neither $\hat{C}_{PL}$ nor $\hat{C}_{PU}$ is an unbiased estimator of the corresponding one-sided index C_{PL} and C_{PU} . The bias can easily be corrected by using an unbiased estimator of $1/\sigma$. The distribution theory for the resulting unbiased one-sided capability indices can be derived

along the same lines as above and such estimators would naturally lead to a new estimator of C_{Pk} , namely the minimum of the two unbiased one-sided indices. The distribution theory for this new estimator can also be worked out. Finally, the conditioning method can be used to find alternative and perhaps less complicated expressions for the sampling distributions of estimators of other capability indices available in the literature.

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